

Boston House Price Prediction

Introduction

House price prediction is an important application of Machine Learning in the real estate industry. It helps buyers, sellers, and investors estimate the value of a property based on different factors such as crime rate, number of rooms, tax rate, and location-related attributes. Accurate house price prediction can support better decision-making and reduce the risks associated with property investments.

The objective of this project is to develop a Machine Learning model that predicts house prices using the Boston Housing dataset. The project involves data preprocessing, exploratory data analysis, model training, and performance evaluation.

Problem Statement

The goal of this project is to predict the median value of houses using various housing-related features available in the dataset. The target variable is MEDV, which represents the median value of owner-occupied homes.

The project aims to build a regression model capable of learning the relationship between housing features and house prices to make accurate predictions.

Dataset Description

The Boston Housing dataset contains 506 records and 14 attributes. These attributes describe different characteristics of houses and their surrounding environment.

Some important features include:

- CRIM: Crime rate by town
- RM: Average number of rooms per dwelling
- AGE: Proportion of older houses
- DIS: Distance to employment centers
- TAX: Property tax rate
- PTRATIO: Student-teacher ratio
- LSTAT: Percentage of lower-status population
- MEDV: Median house value (Target Variable)

The dataset contains some missing values which need to be handled before model training.

Data Preprocessing and Exploratory Data Analysis

The first step of the project was loading the dataset using the Pandas library. The structure and data types of the dataset were examined using the `info()` function.

Missing values were identified using the `isnull()` function. To ensure data quality, missing values were replaced using the mean value of their respective columns.

After preprocessing, exploratory data analysis was performed to better understand the dataset. A correlation heatmap was generated to identify relationships between different variables. Features such as RM and LSTAT showed strong relationships with house prices.

A histogram of the target variable (MEDV) was also plotted to understand the distribution of house prices.

These visualizations helped in understanding the dataset and selecting an appropriate machine learning model.

Model Building

For this project, Linear Regression was selected as the prediction model because it is simple, efficient, and widely used for regression problems.

The dataset was divided into training and testing sets using an 80:20 ratio. The model was trained using the training data and then tested on unseen data.

The following steps were performed:

1. Feature selection
2. Train-test split
3. Model training
4. Prediction generation
5. Performance evaluation

The trained model learned the relationship between housing features and house prices and was able to predict house values for new records.

Results and Evaluation

The model's performance was evaluated using two standard regression metrics:

Mean Squared Error (MSE)

Mean Squared Error measures the average squared difference between actual and predicted values. A lower MSE indicates better performance.

R² Score

The R² Score measures how well the model explains the variation in the target variable. Higher values indicate better predictive performance.

The model achieved an MSE of 25.017672023842703 and an R² Score of 0.658852019550814, indicating good predictive performance.

An Actual vs Predicted scatter plot was generated to visually compare predicted house prices with actual values.

Conclusion

In this project, a Machine Learning model was developed to predict Boston house prices using housing-related features. The project involved data preprocessing, exploratory data analysis, model training, and evaluation.

The Linear Regression model successfully learned the relationship between housing characteristics and house prices and produced satisfactory predictions. This project demonstrates how Machine Learning can be applied to real-world real estate problems and supports data-driven decision-making.